An Uncertainty-Aware Active Learning Framework for Sentiment Analysis Using Transformer Based Models

Submitted by

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1. Introduction

Sentiment analysis is an important task in Natural Language Processing (NLP) used to identify opinions and emotions from text data. It is widely used in social media monitoring, customer feedback analysis, and business intelligence. Traditional techniques such as Bag-of-Words and TF-IDF fail to capture contextual meaning.

To overcome this limitation, transformer-based models such as BERT, RoBERTa, DistilBERT, ALBERT, and DeBERTa are used. However, these models require large labeled datasets, making manual labeling expensive. This project proposes an **uncertainty-aware active learning framework using Expected Gradient Length (EGL)** to improve accuracy while reducing labeling effort.

2. Problem Statement

Existing sentiment analysis models require large labeled datasets and struggle with contextual and implicit sentiment. Manual labeling of large datasets is costly and time-consuming. This project aims to improve classification accuracy while minimizing labeling cost using transformer models with active learning.

3. Objectives

- Implement transformer-based sentiment classification models
- Reduce manual labeling effort using EGL-based active learning
- Improve model performance using contextual embeddings
- Compare performance of BERT, RoBERTa, DistilBERT, ALBERT, and DeBERTa
- Evaluate models using Accuracy, Precision, Recall, F1-score, and AUC

4. Dataset Used

Two datasets are used:

IMDB Movie Reviews Dataset

- 50,000 reviews (25k positive, 25k negative)
- Split: 80% training, 10% validation, 10% testing

Twitter Sentiment Dataset (Augmented)

- 18,944 tweets
- Classes: Positive, Neutral, Negative

Text preprocessing includes cleaning, tokenization, normalization, and padding for transformer input

5. Methodology

1. **Data Preprocessing**
   Cleaning, tokenization, normalization, and sequence padding

2. **Transformer Model Training**
   Fine-tuning BERT, RoBERTa, DistilBERT, ALBERT, and DeBERTa

3. **EGL-Based Active Learning**
   Selecting uncertain samples for labeling to reduce annotation cost

4. **Model Evaluation**
   Using Accuracy, Precision, Recall, F1-score, and ROC-AUC

6. Tools and Technologies

- Python Programming
- NumPy, Pandas, Scikit-learn
- PyTorch / TensorFlow
- HuggingFace Transformers
- Google Colab (GPU support)
- Matplotlib and Seaborn

7. Results (Current Progress)

- DeBERTa achieved highest accuracy (~94%)
- RoBERTa showed strong contextual performance
- EGL Active Learning improved performance and reduced labeling effort
- Transformer models outperformed traditional ML approaches

8. Individual Contribution

M. Shanmukha Priya

- Literature survey
- Model implementation
- Training and evaluation
- Report writing

K. Swathi

- Data preprocessing
- Active learning implementation
- Result analysis
- PPT preparation

9. Work Status

- Literature survey completed
- Dataset collection and preprocessing completed
- Model implementation completed
- Active learning integrated
- Experimental results generated
- Paper draft completed and plagiarism checked

10. Conclusion

The proposed transformer-based sentiment analysis system integrated with EGL Active Learning improves classification accuracy and reduces labeling effort. Among all models, DeBERTa and RoBERTa achieved the best performance, making the system effective for real-world sentiment analysis applications.

11. Future Scope

- Integration of Large Language Models (LLMs)
- Multilingual sentiment analysis
- Sarcasm and emotion detection
- Real-time deployment systems